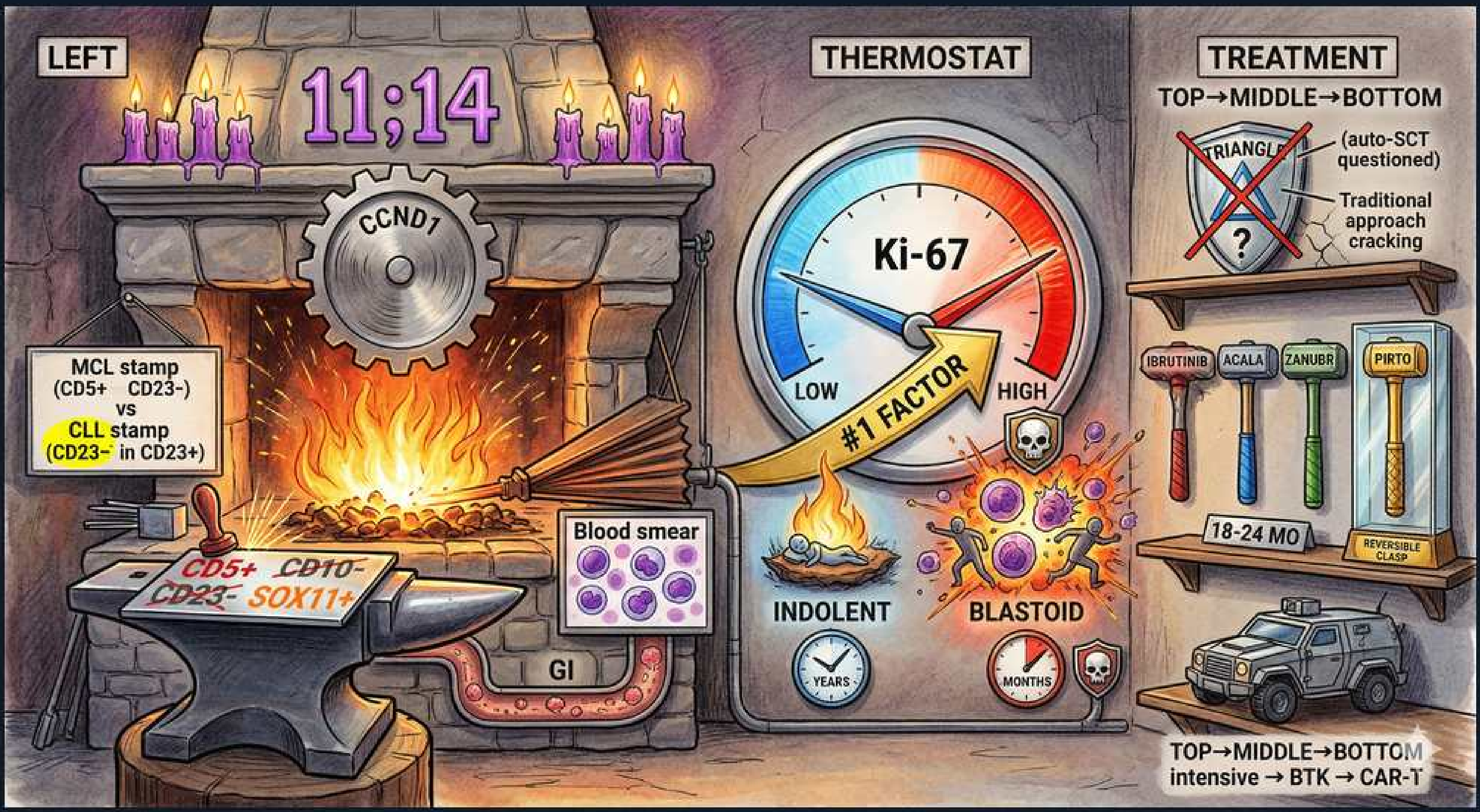
Mantle Cell Lymphoma — Biology, Prognosis & Therapy



1. t(11;14) translocation

WHAT YOU SEE

11;14 above the forge

WHAT IT ENCODES

Mantle cell lymphoma is defined by t(11;14)(q13;q32), juxtaposing CCND1 to IgH. Aggressive yet historically incurable B-NHL — combines aggressive tempo with relapsing indolent-like behavior.

BOARD TRAP

t(11;14) is the MCL signature; t(14;18) is follicular and t(8;14) is Burkitt — don't confuse the partner chromosomes.

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2. Cyclin D1 (CCND1)

WHAT YOU SEE

CCND1 gear

WHAT IT ENCODES

t(11;14) overexpresses cyclin D1, driving G1→S cell-cycle progression. Cyclin D1 positivity on IHC is the diagnostic hallmark, confirmed by t(11;14) FISH.

BOARD TRAP

A CD5+ lymphoma that is cyclin D1+ is mantle cell, not CLL — CLL is cyclin D1-negative.

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3. CD5+ / CD23- / SOX11+

WHAT YOU SEE

Blood smear, CD5+ CD10- CD23- SOX11+

WHAT IT ENCODES

MCL immunophenotype: CD5+, CD20+, CD23-, cyclin D1+, SOX11+, CD10-. SOX11 is positive even in the rare cyclin D1-negative variant, aiding diagnosis.

BOARD TRAP

CD5+/CD23+ = CLL; CD5+/CD23-/cyclin D1+ = MCL. The CD23 status and cyclin D1/SOX11 separate them.

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4. MCL vs CLL stamp

WHAT YOU SEE

MCL stamp (CD5+ CD23-) vs CLL stamp (CD23+)

WHAT IT ENCODES

Both are CD5+ small B-cell neoplasms. CLL: CD23+, dim CD20/sIg, cyclin D1-. MCL: CD23-, cyclin D1+/SOX11+, t(11;14). This fork is heavily tested.

BOARD TRAP

Don't reflexively call a CD5+ small-cell lymphoproliferative disorder CLL — missing MCL changes prognosis and therapy entirely.

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5. GI involvement

WHAT YOU SEE

GI tract from forge

WHAT IT ENCODES

MCL commonly involves the GI tract as lymphomatous polyposis — multiple polyps throughout the colon. Often disseminated (stage IV) at diagnosis with marrow and blood involvement.

BOARD TRAP

Diffuse GI polyposis on colonoscopy can be MCL — biopsy for cyclin D1, don't assume benign polyps.

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6. Ki-67 (#1 prognostic factor)

WHAT YOU SEE

Ki-67 thermostat dial, #1 FACTOR

WHAT IT ENCODES

Ki-67 proliferation index is the single most important prognostic marker in MCL. Low Ki-67 = indolent course (years); high Ki-67 = aggressive, with blastoid/pleomorphic morphology and short survival.

BOARD TRAP

Ki-67, not stage, drives MCL prognosis — a high proliferation index outweighs favorable-looking anatomy.

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7. Indolent variant

WHAT YOU SEE

Low side, INDOLENT, years

WHAT IT ENCODES

A leukemic non-nodal, SOX11-negative indolent MCL variant can be observed initially (watch-and-wait), behaving more like CLL with prolonged survival.

BOARD TRAP

Not all MCL needs immediate treatment — leukemic non-nodal SOX11-negative MCL can be watched, unlike classic nodal MCL.

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8. Blastoid variant

WHAT YOU SEE

High side, BLASTOID, skull, months

WHAT IT ENCODES

Blastoid/pleomorphic MCL has very high Ki-67, often TP53 mutation, aggressive course measured in months, and poor response — a candidate for early CAR-T or clinical trials.

BOARD TRAP

Blastoid morphology + TP53 mutation predicts chemoresistance — standard immunochemotherapy fails; consider CAR-T/novel agents.

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9. BTK inhibitors (relapsed)

WHAT YOU SEE

Shelf: ibrutinib, acala, zanu, pirto

WHAT IT ENCODES

BTK inhibitors — ibrutinib, acalabrutinib, zanubrutinib (covalent), and pirtobrutinib (non-covalent) — are mainstays for relapsed/refractory MCL, overcoming some prior-BTKi resistance.

BOARD TRAP

Pirtobrutinib (non-covalent) can work after covalent-BTKi failure — don't assume all BTK resistance ends BTK-directed therapy.

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10. Intensive first-line ± auto-SCT

WHAT YOU SEE

Top→Middle→Bottom, auto-SCT questioned

WHAT IT ENCODES

Younger fit patients historically received intensive cytarabine-based induction + autologous transplant consolidation; the role of auto-SCT is now questioned with MRD-guided BTKi-containing regimens (e.g., TRIANGLE).

BOARD TRAP

The 'transplant for everyone' MCL paradigm is shifting — adding ibrutinib up front may make auto-SCT unnecessary in some patients.

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11. CAR-T (brexucabtagene)

WHAT YOU SEE

Armored vehicle, intensive→BTK→CAR-T

WHAT IT ENCODES

Brexucabtagene autoleucel (CD19 CAR-T) is approved for relapsed/refractory MCL after BTK inhibitor failure, producing durable remissions in heavily pretreated disease.

BOARD TRAP

After BTKi failure in MCL, CAR-T is the next high-yield option — recognize CRS/ICANS toxicity post-infusion.

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12. 18-24 month BTKi durability

WHAT YOU SEE

18-24 MO vials

WHAT IT ENCODES

Single-agent BTKi responses in relapsed MCL last roughly 18-24 months before progression, framing the need for subsequent CAR-T or combination strategies.

BOARD TRAP

BTKi monotherapy is not curative in MCL — plan the next line (CAR-T) given limited durability.
